

2.1 Decomposition model

ESTP Training
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eurostat 

1 Two-steps models (Tramo-Seats, X12)

- 1 Regression model

- $y_c = X\beta + \mu = Cal \beta_{cal} + Out\beta_{out} + Reg\beta_{reg} + \mu$
 - $\mu \sim ARIMA$ (or another stochastic component)
 - Additive case: $y_c \equiv y - \bar{X}\bar{\beta}$
 - Multiplicative case: $y_c \equiv \ln(y) - \bar{X}\bar{\beta} - \ln(lp)$
- Linearized series: $y_{lin} = y_c - X\beta$

1 Two-steps models

• 2 Decomposition:

- Additive case: $y_{lin} = T_{lin} + S_{lin} + I_{lin}$
- Multiplicative case:
 - $\exp(y_{lin}) = T_{cmp} \times S_{cmp} \times I_{cmp}$ (X11)
 - $\exp(y_{lin}) = \exp(T_{lin}) \times \exp(S_{lin}) \times \exp(I_{lin}) \times (bias\ corr) = T_{cmp} \times S_{cmp} \times I_{cmp}$ (SEATS...)

- Note on bias correction: annual average of $S_{cmp}, I_{cmp} = 1$ (see later)

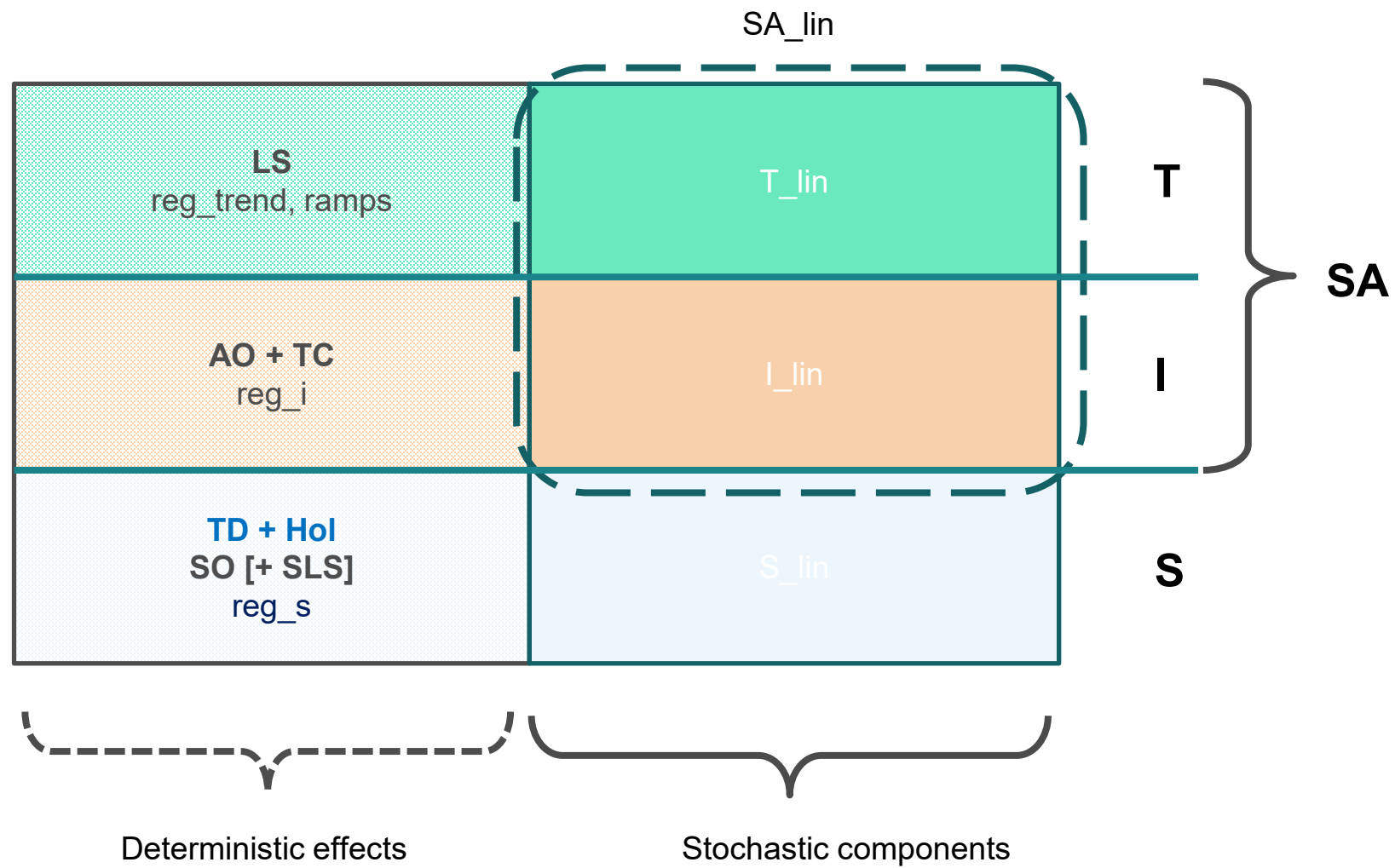
2 Final decomposition

- **Additive case:**

- ✓ $t = t_{lin} + Out_t + Reg_t$
- ✓ $s = s_{lin} + Cal + Out_s + Reg_s$
- ✓ $i = i_{lin} + Out_i + Reg_i$
- ✓ $sa = t + i = y_c - s$

- **Multiplicative case:**

- ✓ $t = t_{cmp} \cdot \exp(Out_t \cdot Reg_t)$
- ✓ $s = s_{cmp} \cdot \exp(lp \cdot Cal \cdot Out_s \cdot Reg_s)$
- ✓ $i = i_{cmp} \cdot \exp(Out_i \cdot Reg_i)$
- ✓ $sa = t \cdot i = y_c / s$



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